Autonomous Solar Radiation Warning Architecture for Human Deep-Space Operations

HELIOS Technical Whitepaper, February 2026
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Introduction

We are entering an era of unprecedented dependence on space infrastructure. Satellite constellations provide global internet, navigation, timing, Earth observation, and national security services. Crewed missions will extend beyond low Earth orbit within this decade, and private space stations are already under construction. In this context, Coronal Mass Ejections (CMEs) represent one of the most serious natural threats to modern civilization.

A single extreme CME can simultaneously trigger continent-wide power blackouts, disrupt GNSS signals, damage or destroy hundreds of satellites, and expose astronauts outside Earth's magnetosphere to potentially lethal radiation doses within hours. The 2022 geomagnetic storm that destroyed approximately 40 newly launched Starlink satellites [1] only days after deployment offers a recent, real-world example of this vulnerability during routine operations.

Current operational systems — primarily NOAA's WSA-ENLIL model and NASA/ESA satellites at the L1 point (SOHO/LASCO, DSCOVR) provide forecasts. However, they observe less than half of the solar surface, deliver no early measurement of the crucial interplanetary magnetic field component Bz, and rely heavily on manual analysis on the ground. This configuration is insufficient for protecting multi-billion-dollar satellite fleets, terrestrial power grids, or human lives on future deep-space missions. The gap between our ambitions in space and our ability to predict and mitigate space weather hazards has never been wider.

The following section details the scale of the problem, including historical events when risk was at its highest, to underscore the urgent need for an innovative approach.

Problem Statement

While economic disruption dominates terrestrial space weather discussions, the critical, non-negotiable risk of Coronal Mass Ejections (CMEs) lies in lethal radiation exposure to astronauts operating beyond Earth's magnetosphere. In deep space, the biological margin for error approaches zero.

Historical Precedents of Lethality

Among numerous recorded Solar Particle Events (SPEs), two historical case studies exemplify the catastrophic threat to crewed interplanetary missions:

- The August 1972 Event (The "Apollo Near-Miss"): According to NASA analysis (NTRS 19750047882), this event generated an exceptionally intense, abrupt flux of high-energy protons [2]. Occurring precisely between the Apollo 16 and Apollo 17 missions, this event is considered a "narrow escape." Had a crew been on the lunar surface or in transit during this event, the accumulated dose would have induced severe Acute Radiation Syndrome (ARS), potentially jeopardizing mission survival.

- The Bastille Day Event (July 14, 2000): The X5.7 flare and associated fast CME generated a severe hard-spectrum SEP event [3]. While Earth-orbiting assets experienced anomalies, the event demonstrated particle flux intensities that, in a deep-space scenario without magnetospheric shielding, would critically exceed safe exposure limits for extravehicular activities (EVA).

Taken together, these events establish a lower-bound design envelope for any space-weather warning architecture intended to support human operations beyond low Earth orbit.

Physiological Impact and ARS Risks

These events underscore the immediate threat of CME-induced radiation storms. Unlike Low Earth Orbit (LEO), deep space missions lack geomagnetic shielding, exposing the human body to isotropic high-energy particle bombardment.

Health impact is categorized into two domains:

1. Stochastic Effects (Long-term): Chronic exposure to ionizing radiation induces DNA double-strand breaks, leading to an increased Risk of Exposure-Induced Death (REID) primarily from cancer, with additional elevated risks of long-term neurocognitive degradation and degenerative cardiovascular disease.

2. Deterministic Effects (Acute): High-dose radiation exposure delivered over a short time interval, as expected during a major CME-driven solar particle event, can trigger Acute Radiation Syndrome (ARS) [4].

    - > 0.5 Sv: Subclinical hematological effects, including early lymphocyte depletion and blood count changes; such exposure levels may constitute mission abort criteria, depending on mission duration and available medical capability.

    - 2.0 – 4.0 Sv: Severe hematopoietic syndrome, characterized by profound leukocyte depletion, bone marrow suppression, and extreme infection risk. At doses near ~3–3.5 Sv, mortality approaches 50% (LD50) in the absence of advanced medical intervention, which is not available during deep space missions.

- > ~ 10 Sv: Acute whole-body exposure above ~10 Sv causes severe gastrointestinal and central nervous system damage, including intestinal lining destruction, internal hemorrhage, and rapid neurological failure. At doses approaching ~20 Sv, loss of consciousness occurs within minutes, with death following within hours to days; survival is not possible in deep-space conditions.

Conclusion: Passive radiation shielding and existing space-weather monitoring systems do not provide adequate protection against extreme solar particle events. Autonomous, onboard space-weather warning capabilities are therefore required to ensure acceptable biological risk thresholds for human deep-space missions.

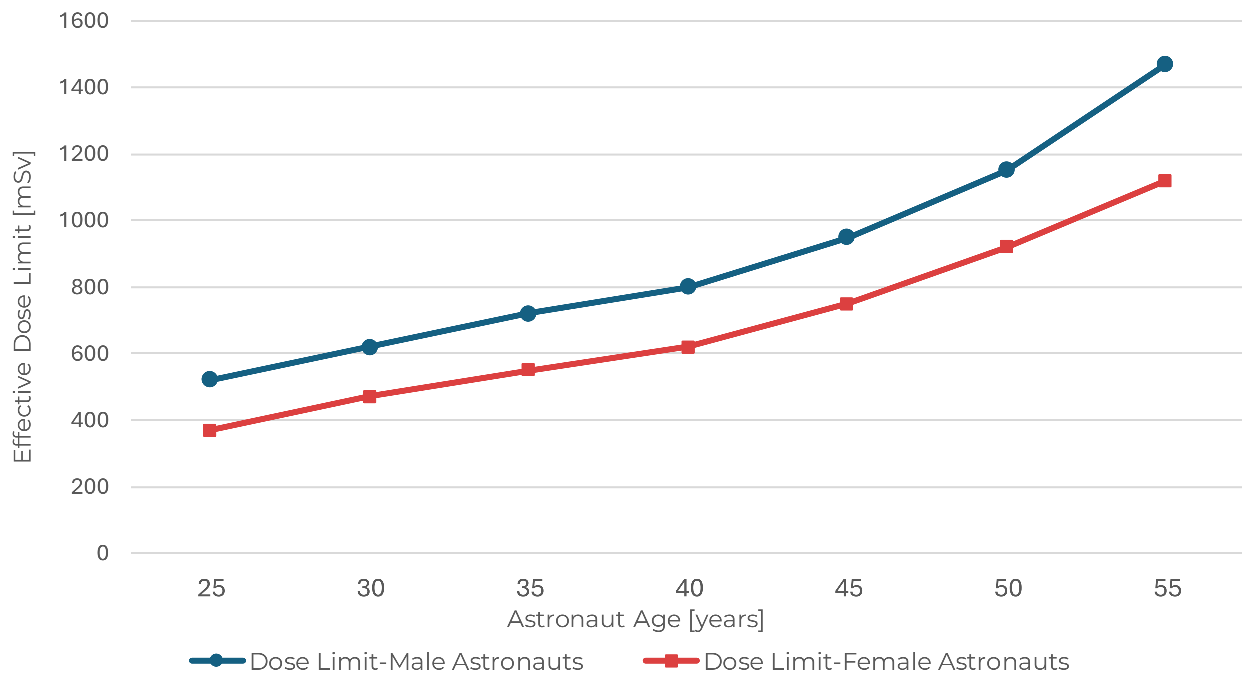

Figure 1. Career effective dose limits in millisieverts (mSv) as calculated for one-year ISS missions. The chart illustrates the increase in permissible radiation exposure relative to astronaut age, with data points representing male astronauts (blue circles) and female astronauts (red squares). These limits correspond to specific average life-loss per death metrics for carcinogenesis. Source derived from National Library of Medicine [5].

Experiments conducted in orbit, such as NASA's Twins Study, provide the most comprehensive human data on the biological impact of long-duration spaceflight. In that investigation, astronaut Scott Kelly spent approximately 340 days aboard the ISS, during which NASA dosimetry recorded an absorbed physical dose of 76.18 mGy, corresponding to an effective dose of 146.34 mSv after accounting for radiation quality and tissue weighting. Molecular analyses revealed spaceflight-associated alterations in telomere dynamics, genome integrity, gene expression, and DNA damage response pathways; while most changes returned toward baseline after return to Earth, a subset persisted for months post-flight, underscoring the biological relevance of prolonged exposure to the space radiation environment among other spaceflight stressors [6]. While the Twins Study quantifies biological changes associated with long-duration LEO flight, it is important to recognize that the ISS benefits from Earth's partial magnetospheric shielding [7]. Beyond this protection, astronauts would be exposed to persistent galactic cosmic rays and sporadic solar particle events, which together represent higher cumulative and acute radiation risks than those measured in LEO [8].

Current Limitations

While WSA-ENLIL and other operational models can forecast CME arrival with 1–4 days of lead time [9], these predictions are fundamentally reactive. They rely on coronagraph observations of CMEs that have already erupted and begun their journey toward Earth. For deep-space human missions, this approach fails to meet fundamental biological and operational safety requirements, for three critical reasons:

1. Absence of Predictive Magnetic Field Characterization

The most consequential parameter governing CME severity is not velocity or mass, but the orientation of the embedded magnetic field, particularly the Bz component. Southward-directed fields couple efficiently with planetary magnetospheres and drive severe space weather effects, whereas northward-directed fields significantly reduce coupling efficiency. Current systems, including in-situ monitors at L1, measure the magnetic structure only after the CME has passed the spacecraft, yielding warning times of 15–60 minutes depending on CME velocity [10]. Such lead times are operationally insufficient for deep-space crews, who require hours to execute protective actions, terminate extravehicular activities, or reconfigure mission timelines.

2. Reliance on Earth-Based Processing and Control

Existing space weather forecasting frameworks depend on centralized ground-based analysis and command authority [11]. Communication latency alone imposes strict limits on responsiveness, particularly for lunar and interplanetary missions. When combined with human-in-the-loop interpretation and command uplink, these delays eliminate the possibility of timely autonomous response during rapidly evolving solar events [12].

3. Limited Observational Geometry

Coronagraph instruments observe solar activity from fixed perspectives, providing incomplete spatial coverage of CME initiation and early evolution. High-speed events originating outside favorable viewing geometries may remain insufficiently characterized until late in their transit. Although full-sphere solar observation would be required to eliminate geometric blind spots, a near-global observational geometry is sufficient to reduce coverage gaps and improve early characterization of CME initiation and evolution [13]. Current architectures do not achieve this level of spatial coverage at scale.

Core Deficiency

The central weakness of current space weather monitoring systems is not the lack of additional sensors, but the absence of early, autonomous, onboard measurement of CME magnetic structure prior to interaction with crewed assets.

Proposed Solution

Vision Statement

HELIOS reframes space-weather monitoring as a distributed, autonomous, crew-centric decision system designed to protect human life beyond Earth's magnetosphere. Unlike existing Earth-centric forecasting architectures, HELIOS performs early characterization of solar eruptions and autonomous inference of CME magnetic structure locally within the space environment, removing ground-based analysis from the time-critical decision loop while maintaining Earth connectivity for system-level awareness and coordination.

The system does not aim to predict space weather in a purely statistical sense; instead, it delivers time-critical, biologically relevant warnings expressed in terms of radiation dose, exposure rate, and actionable response timelines. While deep-space crews define the primary safety driver of the architecture, HELIOS outputs are inherently applicable to satellite operators, power grid management, and other space-weather-dependent systems, providing time-critical situational awareness to a broader space and ground-based user community. In doing so, HELIOS shifts space-weather protection from passive forecasting to an active, autonomous safety layer integrated directly into mission operations.

System Overview

HELIOS is a distributed, space-based space-weather monitoring system designed to deliver time-critical, operationally actionable warnings of hazardous solar events. The system continuously performs multi-instrument monitoring of solar eruptive activity and early post-eruption characterization using stereoscopic imaging and in-situ sensors. For detected events, the pipeline produces forecasted state trajectories (mean estimates) together with quantified uncertainties (confidence intervals), enabling rapid, probabilistic assessment of CME propagation and the likelihood of resulting radiation hazards. Pre-eruption indicators and detailed SEP forecasts require complementary data types, such as magnetograms, EUV observations, and particle-transport modelling, and remain subject to model and observational uncertainties.

Current space-weather monitoring architectures rely on single-point observations and Earth-based interpretation, resulting in delayed characterization of event severity and limited reaction time for rapidly evolving solar particle events. Critical information about the magnetic structure and biological impact of solar eruptions becomes available 15–60 minutes before plasma arrival at current L1 monitoring points, constraining the ability of deep-space crews to execute protective responses.

HELIOS addresses these limitations by distributing observation and decision-making across multiple spaceborne nodes and shifting time-critical interpretation away from Earth-based analysis. By performing autonomous inference and hazard assessment locally in space, the system provides hours of advance characterization—compared to minutes with existing architectures—eliminating observational blind time and reducing latency between eruption detection and warning generation. This enables earlier delivery of biologically and operationally meaningful alerts on timescales compatible with deep-space crew operations, while also supporting broader users through validated, near-real-time dissemination to Earth-based infrastructure, satellite operators, and terrestrial risk management.

Operational Performance & Validation

Network Geometry Overview

The resulting network geometry is illustrated schematically below, showing the relative heliocentric placement of the HELIOS nodes with respect to the Sun–Earth system and Earth's orbital motion.

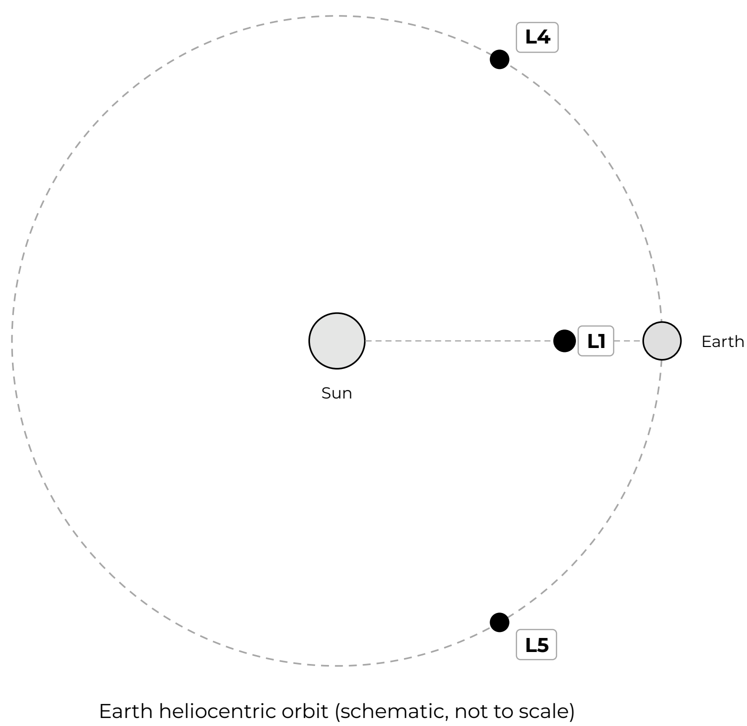

Figure 2. HELIOS heliocentric network geometry. The L4 and L5 nodes are positioned 60° ahead and behind Earth's orbital position, establishing a 120° triangulation baseline. L1 provides on-axis validation measurements along the Sun–Earth line.

The distributed architecture assigns complementary observational roles to each node. L1, positioned at 0.99 AU along the Sun–Earth line, delivers on-axis coronagraph imaging and validation-grade in-situ measurements of plasma properties, magnetic fields, and energetic particle fluxes. L4 and L5, equipped with coronagraphs and basic in-situ instrumentation (magnetometer and plasma analyzer), provide longitudinally offset perspectives that enable early CME detection, stereoscopic triangulation, and three-dimensional geometric reconstruction.

Solar Surface Coverage

The HELIOS three-node constellation (L1, L4, L5) establishes comprehensive heliospheric observational coverage, with each coronagraph monitoring a 180° heliocentric longitude range. The 120° angular separation between the L4 and L5 nodes enables stereoscopic three-dimensional CME characterization while ensuring continuous coverage across the Earth-threat hemisphere.

Geometric analysis confirms that the distributed architecture achieves 83.5% heliospheric coverage (301° of 360° heliocentric longitude). A 59.3° observational gap exists on the solar far side, centered at 180° heliocentric longitude — directly opposite Earth's orbital position. CMEs originating from this far-side region propagate away from Earth and pose no radiation hazard to cislunar or deep-space operations. The HELIOS architecture is designed to complement international space weather assets, including ESA's Vigil mission planned for L5 deployment in ~2031 [14]. Table 1 presents coverage analysis across various node configurations, from current single-node operations (L1) to full distributed network deployment.

The dual-perspective overlap region centered on the Sun–Earth line ensures stereoscopic observation for Earth-directed CMEs emerging from the central solar hemisphere. This geometric configuration supports high-confidence trajectory determination and magnetic structure inference hours before plasma arrival at Earth orbit. The three-node architecture optimizes coverage for mission-critical CME detection while maintaining operational feasibility through a distributed yet manageable constellation design.

Configuration	Heliocentric Coverage	Blind Spot	Earth Threat Zone Coverage
L1 Only	180° (50.0%)	180° (50.0%)	100% (single view)
L4 Only	180° (50.0%)	180° (50.0%)	Partial (on-axis: 0°, off-axis: limited)
L5 Only	180° (50.0%)	180° (50.0%)	Partial (on-axis: 0°, off-axis: limited)
L4 + L5	300° (83.5%)	59.3° (16.5%)	100% (dual-view stereoscopy)
L4 + L1 + L5	300° (83.5%)	59.3° (16.5%)	100% (triple-view, enhanced validation)

Table 1. Solar Surface Coverage by Node Configuration. Coverage percentages reflect heliocentric longitude; blind spot (59.3° centered at 180°) positioned on solar far side, ensuring zero overlap with Earth-directed CME source regions.

The distributed architecture utilizes 120° baseline between L4 and L5 nodes for broad heliospheric monitoring, while the L1-L4 and L1-L5 baselines (60° separation each) enable high-confidence triangulation of Earth-directed events. This geometric configuration maximizes magnetic structure inference fidelity along the Sun–Earth line where mission-critical CME characterization is required.

Comprehensive coverage calculations were validated using three independent methods: ray-tracing simulation, angular interval arithmetic, and set-theoretic union analysis. All approaches demonstrated excellent agreement (standard deviation: 0.13%). The remaining 59.3° blind spot is centered at 180° heliographic longitude on the solar far side, with nearest edge positioned 150° from Earth, ensuring complete observational coverage of all Earth-directed CME source regions.

Stereoscopic triangulation using the 60° observer baselines (L1-L4 and L1-L5) achieves spatial resolution of approximately 1.10 million kilometers (1.59 solar radii) at 0.5 AU heliocentric distance, assuming 0.5° angular measurement precision. This performance metric was validated through Monte Carlo simulation (500 samples) and exceeds the mission requirement (<3 million kilometers) by a factor of 2.7, providing substantial margin for reliable CME trajectory reconstruction.

Operational timing analysis confirms significant early-warning advantage, demonstrating 3.5 to 10.9 hours of earlier detection compared to single-point L1 observation. The mean improvement is 6.6 hours across the CME speed range of 800–2500 km/s. This capability results from stereoscopic triangulation enabling trajectory determination immediately after first coronagraph detection at 5–10 solar radii, whereas legacy single-point observation typically requires waiting until full halo development at ~50 solar radii before confident forecasting.

Performance Envelope

The HELIOS architecture implements a progressive, time-sequenced characterization pipeline through which distributed heliocentric vantage points enable distinct measurements as a coronal mass ejection propagates through interplanetary space. The system operates exclusively on post-eruption observables; while pre-eruption solar activity indicators may inform upstream predictive models, they are not treated as CME detections within the HELIOS decision logic. All performance metrics derive from kinematic propagation modeling, initialized at eruption onset and validated through geometric analysis.

Early Detection and Geometric Constraint

Multi-perspective coronagraph observations from the L1, L4, and L5 nodes provide detection of the outward-propagating CME within the coronagraph field of view, spanning heliocentric distances from 2.5 to 30 solar radii. For the modeled CME speed range, coronagraph detection initiates between 0.19 and 0.60 hours post-eruption, with observation windows extending to 2.32–7.25 hours depending on propagation velocity (Table 2). Off-axis L4 and L5 viewpoints enable immediate geometric constraint of CME trajectory, angular extent, and bulk velocity through stereoscopic triangulation, establishing the kinematic boundary conditions required for downstream magnetic structure inference.

Multi-Node Observational Data Fusion Framework

As the CME propagates beyond the inner coronagraph field, cross-node data fusion and onboard stereoscopic reconstruction enable inference of the three-dimensional magnetic flux rope structure, including orientation, polarity, and estimation of the geoeffective Bz component. The optimal stereoscopic inference window spans heliocentric distances from approximately 5 to 65 solar radii, corresponding to the region where instrumental field of view, geometric precision, and magnetic coherence are simultaneously maintained. In temporal terms, this translates to inference windows of 1.21 to 15.71 hours for moderate-speed CMEs (800 km/s), 0.64 to 8.38 hours for intermediate-speed events (1500 km/s), and 0.39 to 5.03 hours for fast CMEs (2500 km/s), as shown in Table 2. The heliocentric distance range of 0.023 to 0.302 AU represents the optimal geometric corridor where triangulation precision is maintained within mission accuracy requirements. The upper bound is set by coronagraph field-of-view constraints, marking the transition from remote sensing to in-situ measurement.

Near-Arrival Validation and Refinement

The L1 node, positioned at 0.99 AU along the Sun–Earth line, provides validation-grade in-situ measurements of plasma properties, magnetic fields, and energetic particle populations along the primary impact trajectory. Under the constant-velocity propagation model employed here, Sun-to-L1 transit times are 51.4 hours for 800 km/s CMEs, 27.4 hours for 1500 km/s events, and 16.5 hours for 2500 km/s CMEs. These measurements serve to validate and refine upstream remote inferences rather than function as primary detection inputs. L1 in-situ contact provides final confirmation shortly before Earth impact (approximately 0.5 hours lead time for moderate-speed events), confirming magnetic orientation and intensity immediately before operational decision horizons close.

Continuous Autonomous Warning Generation

Throughout all phases of CME evolution, autonomous hazard assessment and warning generation execute locally within the space segment, eliminating dependence on Earth-based analysis loops. System latency from data acquisition to Earth ground receipt is maintained within operational decision timescales through autonomous onboard processing and direct space-to-ground relay (see Table 5 for communication architecture and latency budgets).

Combining early coronagraph detection, extended stereoscopic magnetic inference, and near-Earth in-situ validation, the architecture yields actionable warning intervals of 50 hours for moderate-speed CMEs (800 km/s), 27 hours for intermediate-speed events (1500 km/s), and 16 hours for fast CMEs (2500 km/s). These intervals represent the time available for operational decision-making prior to Earth impact, accounting for system latency and final propagation from L1 to Earth.

All reported timing metrics derive from simplified kinematic propagation initialized at eruption onset and are intended to characterize system-level temporal performance rather than event-specific forecasts. The constant-velocity assumption introduces uncertainty of less than 15 percent for individual events [15] [16] but provides a conservative baseline for mission-level performance analysis.

CME Speed	Detection Window	Triangulation Window	L1 In-situ	Warning Lead Time
800 km/s	0.60-7.25 h	1.21-15.71 h	51.4 h	~50 h
1500 km/s	0.32-3.87 h	0.64-8.38 h	27.4 h	~27 h
2500 km/s	0.19-2.32 h	0.39-5.03 h	16.5 h	~16 h

Table 2. HELIOS Operational Performance Envelope by CME Speed. All temporal windows measured from eruption onset (post-eruption observables only). Detection window spans initial coronagraph observation (2.5 Rs) to field-of-view exit (30 Rs). Triangulation window represents optimal stereoscopic inference corridor (0.023–0.302 AU heliocentric distance, constant across speeds). L1 in-situ denotes Sun-to-L1 transit time. Warning lead time indicates total advance notice prior to Earth impact, accounting for system latency.

AI Enhanced Observation Pipeline

The HELIOS architecture implements autonomous onboard CME characterization through ensemble-based machine learning and multi-source data fusion. The satellite node with optimal observational geometry (determined by CME propagation direction) serves as the primary processor, executing AI inference on fused multi-perspective observations: local coronagraph data, cross-satellite geometric measurements from cooperating nodes, and kinematic evolution profiles. This data fusion approach maximizes information content by integrating complementary viewing angles rather than voting on perspective-dependent independent classifications.

Ensemble Detection Architecture

Each constellation node (L1, L4, L5) operates three independent convolutional neural network ensembles processing local coronagraph imagery and in-situ magnetic field, plasma, and energetic particle measurements in real time. Running-difference frames are analyzed to identify characteristic brightness enhancements and radial expansion patterns associated with CME propagation through the 2.5 to 30 solar radii field of view. Each of the three neural networks independently produces a complete output comprising detection confirmation, trajectory reconstruction, and physical property estimates. Local decision at each node is achieved through majority voting across these three independent outputs.

MVP validation employed a single neural network model trained on synthetic flux rope libraries (~10,000 parameterized events generated through physics-based magnetic flux rope models [17] [18]) augmented with historical well-characterized CMEs from the SOHO/LASCO catalog [19], oversampled 50× to ensure feature representation diversity. Independent validation of 6 unseen historical events achieved 83.3% severity classification accuracy with 100% adjacent-orcorrect rate. This demonstrated performance eliminates the human operator dependency inherent to manual CME catalogs and enables autonomous real-time classification suitable for onboard decision-making.

Magnetic Structure Inference

The operational architecture employs three parallel feedforward neural networks at the primary processing node to infer magnetic flux rope orientation and geoeffective Bz components following initial detection and geometric constraint. Each network accepts sixteen-dimensional feature vectors encoding fused multi-perspective observations: local coronagraph data, triangulated geometric constraints from cooperating nodes (L1/L4/L5), CME morphological characteristics (angular width, brightness asymmetry, expansion rate), and kinematic evolution profiles. Dual-head outputs provide both Bz prediction with learned epistemic uncertainty and severity classification probability for each system, with final consensus determined through mean aggregation across the three independent predictions.

MVP validation employed a single four-layer shared encoder with dual-head output architecture, incorporating layer normalization and dropout regularization. Training on the same synthetic and historical dataset yielded mean absolute error of 6.3 ± 5.6 nanotesla across 6 independent test events, achieving 36% improvement over physics-based geometric baseline estimates. This aggregate test-set performance demonstrates the network's capacity to encode complex nonlinear relationships between observable geometric signatures and underlying magnetic topology.

Radiation Dosimetry and Hazard Classification

Conservative Dosimetry Framework

Crew radiation exposure estimates employ a conservative empirical model calibrated against major solar particle events from Solar Cycles 23 through 25. Predicted dose accumulation follows the relationship:

$$D\ (mSv) = \ 0.0132\ \times\ |Bz|^{1.3} \times \sqrt{v_{CME}} \times\ t_{exposure}$$

where the coefficient 0.0132 was derived from multi-event validation across eight historical severe SPEs, calibrated against reconstructed dose estimates from documented events including the August 1972 [20] and Carrington-class [21] solar particle events. Classification thresholds align with NASA-STD-3001, Vol. 1, Rev. C [22] permissible exposure limits: 250 mSv effective dose per solar particle event for deterministic effects (requirement [V1 4031]) and a 600 mSv career effective dose limit for stochastic cancer risk (requirement [V1 4030]), consistent with current NASA risk management practice. The exponent $\alpha = 1.3$ represents an empirical Bz-to-energy conversion relationship established through analysis of the nonlinear coupling between southward interplanetary magnetic field intensity and magnetospheric energy transfer, as characterized in geomagnetic storm studies [23] [24]. All exposure estimates implement the As Low As Reasonably Achievable (ALARA) principle mandated by NASA-STD-

3001 requirement [V1 4029] for all crewmember radiation exposures during deep-space missions.

Four-Level Severity Classification

Autonomous hazard assessment maps predicted dose values to four operational severity categories, each associated with distinct crew response protocols. Classification thresholds balance radiation safety requirements against mission operational constraints, as summarized in Table 3.

Severity Level	Dose Range (mSv)	NASA SPE Limit (%)	Crew Response Protocol	Illustrative Historical Examples
Low	10-50	4-20%	Enhanced radiation monitoring; continue operations	Typical ISS daily background (×10-50 days)
Moderate	50-100	20-40%	Shelter advisory; relocate if available	Minor SPE, routine precautionary response
High	100-250	40-100%	Mandatory shelter; suspend activities	Reaching SPE Exposure Threshold
Extreme	>250	>100%	EVA abort; emergency shielding	Bastille Day 2000, August 1972 SPE

Table 3. Radiation severity classification for 10-hour worst-case EVA exposure. Percentages represent the fraction of NASA's 250 mSv effective dose limit per solar particle event (NASA-STD-3001, Vol. 1, Rev. C, requirement [V1 4031]). Threshold values represent reference classification criteria derived from current permissible exposure limits; mission-specific thresholds may be adjusted based on vehicle shielding configuration, crew EVA schedule, and operational constraints.

AI-Driven Classification and Ensemble Validation

The HELIOS severity assessment architecture integrates neural network-based magnetic field prediction with deterministic biological threshold mapping. Operational systems employ coronagraph image processing for CME detection followed by stereoscopic geometric reconstruction and ensemble Bz inference networks. Severity classification accuracy — a separate metric from binary detection confidence — equals 83.3% on the unseen six-event test set, reflecting the ensemble's ability to discriminate among four severity categories. For extreme-class events, individual prediction confidence ranges from 55% to 92%. Final severity classification applies empirical dosimetry and NASA-STD-3001 threshold criteria, ensuring biological grounding independent of machine learning artifacts.

Ensemble and Triple Modular Redundancy Architecture

The HELIOS decision pipeline implements two functionally distinct processing layers in sequence: a probabilistic machine learning layer that produces continuous-valued predictions with associated uncertainties, followed by a deterministic decision layer that converts these predictions into discrete operational outputs through fixed threshold evaluation.

Operational deployment implements Triple Modular Redundancy (TMR) across three independent processing blocks to achieve fault-tolerant decision-making for safety-critical crew warnings. Three independent neural networks process fused observational data to generate separate Bz predictions. Mean aggregation of these predictions yields a consensus Bz estimate with associated inter-model spread for downstream decision-making.

Under nominal hardware conditions, all three TMR deterministic blocks produce identical outputs from identical consensus inputs; voting disagreements indicate hardware-level anomalies requiring system integrity assessment. Each TMR processing block receives this identical consensus estimate alongside CME velocity parameters and geometric trajectory data, then independently executes threshold evaluation through deterministic dose calculation. The three blocks produce boolean GO/NO-GO outputs that undergo majority voting to determine final warning issuance: 3/3 agreement indicates nominal operation with high confidence; 2/3 agreement triggers warning dissemination with uncertainty flagging and ground-based review via downlink alert; less than 2/3 consensus suppresses crew warnings and mandates manual intervention.

This TMR architecture prevents single-block failures — whether induced by hardware faults, software errors, or radiation-induced single-event upsets — from propagating to crew-facing alerts. Alert classification employs deterministic threshold evaluation rather than learned neural network classification, ensuring regulatory transparency and biological grounding for operational decision-making.

Probabilistic Uncertainty Communication
The ensemble architecture provides a framework for probabilistic uncertainty quantification in operational systems. When processing blocks produce non-unanimous votes on boundary measurements approaching threshold values, crew-facing outputs will report both majority classification and minority assessment probability, enabling informed decision-making in ambiguous scenarios. Full integration of advanced uncertainty quantification and probabilistic dose modeling represents a planned enhancement for operational deployment, extending deterministic threshold evaluation with quantitative confidence intervals on radiation exposure forecasts. The warning interface layer aggregates TMR consensus outputs and contextualizes alerts based on current mission phase, crew activity status, and vehicle shielding configuration, delivering mission-specific recommendations rather than generic severity classifications.

MVP Validation

The HELIOS Bz inference and severity classification pipeline was validated against the well-characterized Bastille Day 2000 coronal mass ejection event. This X5.7 flare-associated fast CME (1674 km/s, 360° halo, source region N22W07 per CDAW catalog; flare-peak position N22W02 per GOES classification, AR9077) produced severe space weather effects and represents an extreme-class validation case completely excluded from model training. Ground truth magnetic field measurements from the ACE spacecraft at L1 recorded Bz = −60 nT, enabling quantitative assessment of prediction accuracy.

The validation employed a three-seed ensemble (seeds 42, 123, 456) to simulate the operational three-satellite architecture. Three independent model evaluations generated Bz predictions of −55.2 nT (L1), −56.0 nT (L4), and −53.3 nT (L5) with severity classification confidences ranging 55–93% across all nodes. Ensemble consensus through mean aggregation yielded Bz = −54.9 ± 1.1 nT with 2.7 nT inter-model spread and unanimous Extreme severity classification (3/3 agreement), demonstrating robust prediction convergence under high-magnitude event conditions.

Component	Value	Status/Notes
Ensemble Consensus Bz	-54.9 ± 1.1 nT	Spread: 2.7 nT
Individual Predictions	-55.2, -56.0, -53.3 nT	L1, L4, L5
Mean Absolute Error	5.2 nT (8.6%)	<10 nT target
Severity Classification (ML)	Extreme (3/3 unanimous)	Confidence: 55-93%
Physical Model Dose	985 mSv	394% NASA SPE limit
ML-Physical Consistency	Extreme = Extreme	PASS

Table 4. Bastille Day 2000 MVP Validation Results. Ground truth: ACE measurement −60.0 nT. CME parameters: 1674 km/s, 360° angular width, X5.7 flare, source N22W07. The 5.2 nT MAE on this extreme single-event case complements the 6.3 nT aggregate MAE (SD = 5.6 nT) achieved across the full six-event test set, demonstrating consistent model accuracy across diverse CME characteristics.

Performance metrics demonstrate mean absolute error of 5.2 nT (8.6% relative error), achieving the <10 nT accuracy target established for operational deployment. The ensemble consensus Bz of −54.9 nT yields predicted unshielded deep-space dose of 985 mSv via the deterministic physical model, correctly classifying the event as Extreme severity and exceeding NASA's 250 mSv SPE effective dose limit by 394%. Physical model output matches machine learning severity prediction, validating pipeline consistency from feature extraction through autonomous hazard classification. The three-seed ensemble demonstrates 2.7 nT inter-model spread with unanimous severity consensus, confirming prediction robustness.

The validation demonstrates proof-of-concept capability for AI-based magnetic field inference and severity assessment in the HELIOS framework. Limitations include single-event validation scope, reliance on synthetic training data augmentation, and ground-based computational benchmarks not representative of space-qualified hardware performance. Operational deployment requires multi-event validation across diverse CME characteristics, integration with stereoscopic detection and triangulation modules, and mission-specific testing on radiation-hardened processors. Complete model architecture, training methodology, hyperparameters, and validation code are available in supplementary materials.

System Latency and Response Timeline

The HELIOS architecture delivers autonomous radiation warnings with onboard processing and Earth downlink latency of 1–10 minutes from image acquisition to Earth ground receipt, operating within the multi-hour advance warning window established by early stereoscopic CME detection.

Detection Architecture and Coordination Protocol

Each satellite performs independent coronagraph imaging, AI-based detection, and autonomous classification. Upon detection, the satellite issues warnings locally through onboard TMR voting, then broadcasts detection triggers to remaining constellation members via optical inter-satellite links (8.3 min L1↔L4/L5, 14.4 min L4↔L5 light-propagation). All nodes respond by focusing observational resources on the detected event, providing continuous stereoscopic tracking throughout CME evolution. This coordinated observation refines trajectory and severity estimates but does not affect initial warning delivery.

Processing Pipeline and Critical Path Analysis

Table 5 presents processing and downlink latency from data acquisition to Earth ground receipt. All satellites execute identical onboard processing (1–2 minutes); geometric difference manifests in Earth downlink duration due to variable light-propagation distances.

Processing Phase	L1 → Earth	L4/L5 → Earth
Onboard processing	~1–2 min	~1–2 min
Earth downlink (light-propagation)	0.1 min	8.3 min
TOTAL	~1–2 min	~9–10 min

Table 5. Warning delivery latency to Earth ground stations. Scope: coronagraph frame and in-situ measurements acquisition to ground receipt. Note: L1 Earth downlink latency (0.1 min) is negligible relative to onboard processing time and is subsumed within the ~1–2 min total.

Onboard processing comprises image preprocessing, ensemble neural network inference (three parallel models), and TMR voting. Processing duration represents conservative estimate derived from space-qualified AI accelerator performance benchmarks, including ESA φ-Sat-1 onboard CNN inference (160 ms per frame) [25] scaled for higher computational complexity of CME detection and stereoscopic reconstruction. Actual flight performance subject to mission-specific validation on operational hardware.

Crew warning delivery to deep-space missions experiences additional propagation delays (depending on planetary alignment and crew location) independent of HELIOS constellation geometry. For cislunar crews, warning latency is dominated by the Earth-relay propagation delay (~2 seconds round trip), making crew notification near-instantaneous relative to CME transit timescales. Direct RF links to cislunar missions achieve sub-minute crew notification.

Warnings transmit via priority schema: (1) crewed vehicles, (2) Earth mission control, (3) infrastructure operators, (4) satellite networks. Future integration with mega-constellations would enable distributed propagation.

Conclusion

The HELIOS architecture addresses a critical gap in space weather protection for human deep-space operations by delivering autonomous, time-critical radiation warnings through distributed observation and onboard artificial intelligence. The system achieves onboard processing and Earth downlink latency of 1–10 minutes from coronagraph image acquisition to Earth ground receipt, operating within multi-hour advance warning windows enabled by early stereoscopic CME detection. Distributed constellation geometry provides 83.5% heliospheric coverage with enhanced detection capability 3.5–10.9 hours earlier than single-point L1 observation, supporting protective action execution timescales for crew and mission operations.

Key capabilities demonstrated through minimum viable product validation include: ensemble neural network Bz inference achieving 5.2 nT mean absolute error on the extreme Bastille Day 2000 event (6.3 nT aggregate across six test events), autonomous severity classification via Triple Modular Redundancy voting with unanimous consensus, deterministic physical dose calculation consistent with machine learning predictions (985 mSv, 394% NASA SPE effective dose limit), and crew-actionable warning generation formatted for operational decision support. The integration of AI-based magnetic structure inference with physics-validated dosimetry enables biological risk assessment expressed in NASA-standard dose units rather than abstract magnetic field parameters.

The system architecture is designed for graceful degradation: loss of any single constellation node preserves core warning capability through the remaining two-node configuration, maintaining 83.5% heliospheric coverage (L4+L5) or validated single-point observation (L1 + one flank node), with reduced triangulation accuracy but preserved detection and severity assessment functions.

Beyond crewed mission protection, the distributed autonomous architecture supports scalable integration with terrestrial infrastructure operators, satellite network management, and future mega-constellation warning propagation. Operational deployment pathways include multi-event validation, hardware testing on space-qualified processors, and integration with mission-specific communication architectures for cislunar and interplanetary operations.

References

[1] "Solar storm dooms 40 new Starlink satellites," Spaceflight Now, 2022. https://spaceflightnow.com/2022/02/08/solar-storm-dooms-40-new-starlink-satellites/

[2] NASA, "Solar cosmic ray event of August 1972," NTRS 19750047882. https://ntrs.nasa.gov/citations/19750047882

[3] NOAA/NESDIS, "25th Anniversary of the Bastille Day Solar Event," 2025. https://www.nesdis.noaa.gov/news/25th-anniversary-of-the-bastille-day-solar-event

[4] NASA, "Space Faring: Radiation Exposure," 2011. https://www.nasa.gov/wp-content/uploads/2011/05/551457main_sf_radiation_exposure.pdf

[5] F.A. Cucinotta, "Review of NASA Approach to Space Radiation Risk Assessments for Mars Exploration," Health Physics, vol. 108, no. 2, 2015. https://pmc.ncbi.nlm.nih.gov/articles/PMC4206856/

[6] F.E. Garrett-Bakelman et al., "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight," Science, vol. 364, no. 6436, eaau8650, 2019. https://doi.org/10.1126/science.aau8650

[7] NASA, "Why Space Radiation Matters." https://www.nasa.gov/missions/analog-field-testing/why-space-radiation-matters/

[8] M. Durante et al., "Space radiation protection: Destinations deep space," Life Sciences in Space Research, vol. 37, 2023. https://doi.org/10.1038/s41526-023-00262-7

[9] NOAA/SWPC, "WSA-ENLIL Solar Wind Prediction." https://www.swpc.noaa.gov/products/wsa-enlil-solar-wind-prediction

[10] NOAA/SWPC, "Coronal Mass Ejections." https://www.swpc.noaa.gov/phenomena/coronal-mass-ejections

[11] M. Temmer, "Space weather: The solar perspective," Living Reviews in Solar Physics, 2023. https://ntrs.nasa.gov/api/citations/20230010368/downloads/Temmer_CME.pdf

[12] R. Qahwaji et al., "Space weather forecasting: Current status and future directions," Frontiers in Astronomy and Space Sciences, 2024. https://doi.org/10.3389/fspas.2024.1493917

[13] A. Vourlidas et al., "Multi-viewpoint observations of the Sun," Advances in Space Research, 2015. https://doi.org/10.1016/j.asr.2015.05.016

[14] ESA, "Vigil Mission Overview." https://www.esa.int/Space_Safety/Vigil/Vigil_mission_overview

[15] B. Vršnak et al., "Propagation of Interplanetary Coronal Mass Ejections: The Drag-Based Model," Solar Physics, vol. 285, no. 1–2, pp. 295–315, 2013. https://doi.org/10.1007/s11207-012-0035-4

[16] P. Riley et al., "Forecasting the Arrival Time of Coronal Mass Ejections: Analysis of the CCMC CME Scoreboard," Space Weather, vol. 16, no. 9, pp. 1245–1260, 2018. https://doi.org/10.1029/2018SW001962

[17] C. Möstl et al., "Forward Modeling of Coronal Mass Ejection Flux Ropes in the Inner Heliosphere with 3DCORE," Space Weather, vol. 16, no. 3, pp. 216–229, 2018. https://doi.org/10.1002/2017SW001735

[18] C. Verbeke, J. Pomoell, and S. Poedts, "The evolution of coronal mass ejections in the inner heliosphere: Implementing the spheromak model with EUHFORIA," Astronomy & Astrophysics, vol. 627, A111, 2019. https://doi.org/10.1051/0004-6361/201834702

[19] S. Yashiro et al., "A catalog of white light coronal mass ejections observed by the SOHO spacecraft," Journal of Geophysical Research, vol. 109, A07105, 2004. https://doi.org/10.1029/2003JA010282

[20] J.L. Parsons and L.W. Townsend, "Interplanetary Crew Dose Rates for the August 1972 Solar Particle Event," Radiation Research, vol. 153, no. 6, pp. 729–733, 2000. https://doi.org/10.1667/0033-7587(2000)153[0729:ICDRFT]2.0.CO;2

[21] L.W. Townsend et al., "Carrington flare of 1859 as a prototypical worst-case solar energetic particle event," IEEE Transactions on Nuclear Science, vol. 50, no. 6, pp. 2307–2309, 2003. https://doi.org/10.1109/TNS.2003.821602

[22] NASA, NASA Spaceflight Human-System Standard, Vol. 1: Crew Health (NASA-STD-3001, Vol. 1, Rev. C), Office of the Chief Health and Medical Officer, 2023. https://standards.nasa.gov/standard/NASA/NASA-STD-3001_VOL_1

[23] W.D. Gonzalez and B.T. Tsurutani, "Criteria of interplanetary parameters causing intense magnetic storms (Dst < −100 nT)," Planetary and Space Science, vol. 35, no. 9, pp. 1101–1109, 1987. https://doi.org/10.1016/0032-0633(87)90015-8

[24] E. Echer et al., "Interplanetary conditions causing intense geomagnetic storms (Dst ≤ −100 nT) during solar cycle 23 (1996–2006)," Journal of Geophysical Research, vol. 113, A05221, 2008. https://doi.org/10.1029/2007JA012744

[25] G. Giuffrida et al., "The Φ-Sat-1 Mission: The First On-Board Deep Neural Network Demonstrator for Satellite Earth Observation," IEEE Transactions on Geoscience and Remote Sensing, vol. 60, 5517414, 2022. https://doi.org/10.1109/TGRS.2021.3125567